

DATABASE DESIGN DOCUMENT

LendWise – MSME Financial Risk Management System

1. Introduction

The database design is a crucial component of the **LendWise system**, as it ensures structured storage, efficient retrieval, and secure handling of MSME financial, transactional, and risk-related data.

This database is designed based on the **Software Requirements Specification (SRS)** and supports core functionalities such as:

- MSME user management
- Financial data processing
- Credit scoring and risk prediction
- Fraud detection
- Dashboard analytics
- Audit logging and compliance

Purpose of Database Design

- Ensure proper data organization and integrity
- Support efficient CRUD operations
- Enable scalability with growing MSME data
- Provide secure storage for financial and sensitive business data
- Maintain audit logs for regulatory compliance

The database forms the **data layer** of the three-tier architecture and interacts with backend services and AI modules.

2. Database Overview

LendWise uses a **Relational Database Management System (RDBMS)** (e.g., MySQL/PostgreSQL).

Key Features

- Structured storage of MSME business data
- Integration with banking systems and transaction sources
- Secure storage with encryption support
- ACID compliance (Atomicity, Consistency, Isolation, Durability)
- Supports analytics, reporting, and AI models

3. Database Entities and Tables

3.1 Users Table

Stores all system users (loan officers, analysts, admin).

Field Name	Data Type	Description
user_id	INT (PK)	Unique user ID
name	VARCHAR	Full name
email	VARCHAR (UNIQUE)	Email address
password	VARCHAR	Encrypted password
role	VARCHAR	Role (Loan Officer, Analyst, Admin)
contact_info	VARCHAR	Contact details
created_at	DATETIME	Registration timestamp

Description:

Each user must have a unique email. Role-based access is controlled here.

3.2 MSME_Details Table

Stores MSME business profile information.

Field Name	Data Type	Description
msme_id	INT (PK)	Unique MSME ID
user_id	INT (FK)	Linked user
business_name	VARCHAR	Name of business
business_type	VARCHAR	Industry type
gst_number	VARCHAR	GST registration
registration_date	DATE	Business registration date
address	TEXT	Business address

Description:

Each MSME is associated with a user.

3.3 Financial_Records Table

Stores summarized financial data used for credit scoring.

Field Name	Data Type	Description
record_id	INT (PK)	Unique record ID
msme_id	INT (FK)	MSME reference
monthly_revenue	DECIMAL	Monthly revenue
monthly_expenses	DECIMAL	Monthly expenses
profit	DECIMAL	Net profit
gst_filing_status	VARCHAR	Filed/Not Filed
created_at	DATETIME	Timestamp

Description:

Used as input for AI-based credit scoring.

3.4 Transactions Table

Stores transaction-level business activity.

Field Name	Data Type	Description
transaction_id	INT (PK)	Unique transaction ID
msme_id	INT (FK)	MSME reference
transaction_date	DATE	Date of transaction
amount	DECIMAL	Transaction amount
type	VARCHAR	Credit/Debit
description	TEXT	Transaction details

Description:

Used for behavioral analysis and fraud detection.

3.5 Credit_Scores Table

Stores generated credit scores and predictions.

Field Name	Data Type	Description
score_id	INT (PK)	Unique score ID
msme_id	INT (FK)	MSME reference
credit_score	INT	Predicted score
risk_level	VARCHAR	Low/Medium/High
model_version	VARCHAR	AI model version
generated_at	DATETIME	Timestamp

Description:

Stores explainable AI results for auditing and decision-making.

3.6 Fraud_Alerts Table

Stores detected fraud activities.

Field Name	Data Type	Description
alert_id	INT (PK)	Unique alert ID
transaction_id	INT (FK)	Related transaction
risk_flag	VARCHAR	Suspicious/High risk
description	TEXT	Alert details
created_at	DATETIME	Timestamp

Description:

Supports fraud detection system.

3.7 Reports Table

Stores analytics and downloadable reports.

Field Name	Data Type	Description
report_id	INT (PK)	Unique report ID
msme_id	INT (FK)	MSME reference
report_type	VARCHAR	Credit/Fraud/Analytics
generated_by	INT (FK)	User ID
created_at	DATETIME	Timestamp

3.8 Audit_Log Table

Tracks all system activities.

Field Name	Data Type	Description
log_id	INT (PK)	Unique ID
user_id	INT (FK)	Acting user
action	VARCHAR	Action performed
details	TEXT	Description

timestamp	DATETIME	Time of action
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3.9 Admin Table

Stores admin-specific data.

Field Name	Data Type	Description
admin_id	INT (PK)	Admin ID
user_id	INT (FK)	Linked user
privileges	TEXT	Permissions

4. Relationships Between Tables

4.1 One-to-Many Relationships

- One User → Many MSMEs
- One MSME → Many Financial Records
- One MSME → Many Transactions
- One MSME → Many Credit Scores
- One MSME → Many Reports
- One User → Many Audit Logs

4.2 One-to-One / Optional

- MSME → Credit Score (latest)

4.3 Relationship Summary

- Users manage MSMEs
- MSMEs generate financial and transaction data
- Credit scores are calculated per MSME
- Fraud alerts are linked to transactions
- Reports summarize all data

5. ER Design Summary

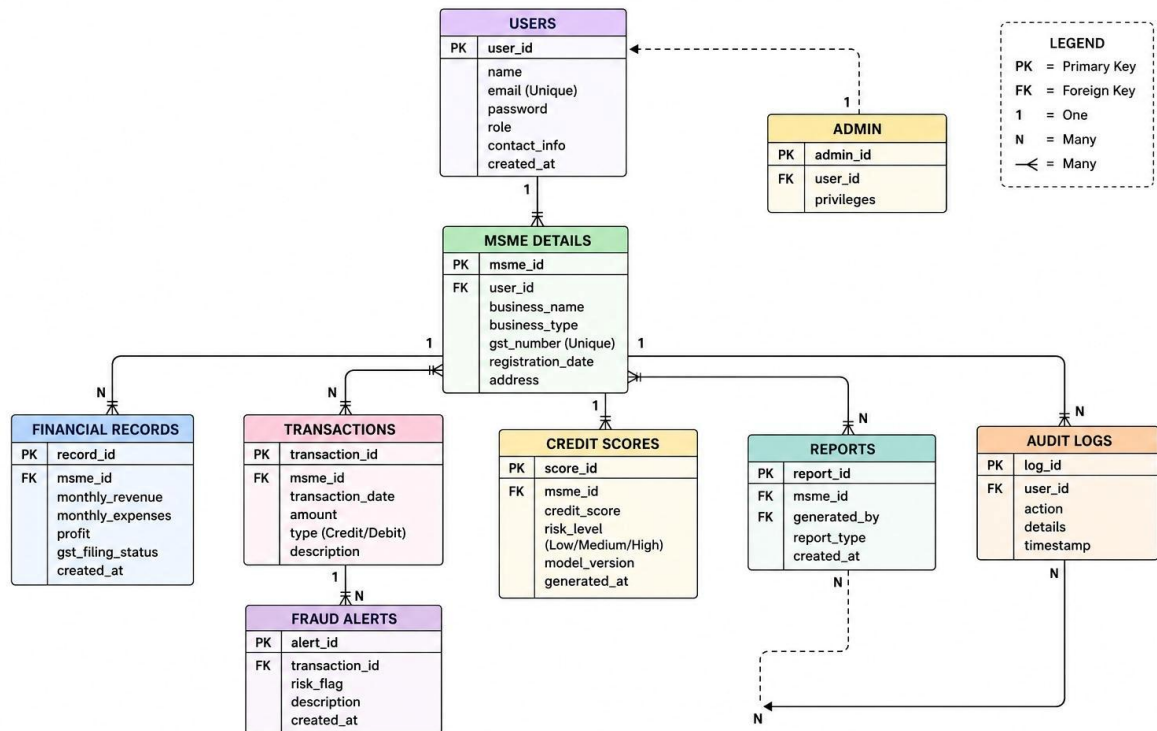
Entities:

- User
- MSME
- FinancialRecord
- Transaction
- CreditScore
- FraudAlert
- Report
- Admin
- AuditLog

Key Relationships:

- User → MSME (1:N)
- MSME → Transactions (1:N)
- MSME → Financial Records (1:N)
- MSME → Credit Score (1:N)
- Transaction → FraudAlert (1:N)

ER DIAGRAM – MSME FINANCIAL RISK MANAGEMENT SYSTEM



6. Constraints

6.1 Primary Keys

- user_id, msme_id, transaction_id, score_id

6.2 Foreign Keys

- user_id in MSME
- msme_id in Transactions, Scores, Records

6.3 Unique Constraints

- Email must be unique
- GST number must be unique

6.4 Not Null Constraints

- Required business and financial fields

6.5 Data Validation

- Risk levels: Low / Medium / High
- Transaction type: Credit / Debit

7. Security Considerations

- Password hashing and encryption
- Role-based access control
- Secure APIs (HTTPS)
- Data encryption at rest and in transit
- Audit logs for monitoring
- SQL injection prevention

8. Performance Considerations

- Indexing on:
 - msme_id
 - transaction_id
 - user_id
- Query optimization for analytics
- Real-time fraud detection support
- Backup and recovery system
- Scalable architecture for large datasets

9. Future Enhancements

- Integration with GST APIs
- External banking data integration
- Advanced ML models for prediction
- Real-time analytics dashboards
- Multi-language support

10. Conclusion

The LendWise database provides a **robust, scalable, and secure foundation** for MSME risk management. It supports:

- AI-based credit scoring
- Fraud detection
- Real-time analytics
- Regulatory compliance

The modular design ensures easy maintenance and future expansion, making it suitable for modern banking systems.